Note 1.5 Elementary Matrices

Equivalent Systems

- If an $m \times n$ linear system $A\mathbf{x} = \mathbf{b}$, then if we multiply both sides of the equation by a nonsingular $m \times m$ matrix M, then the new system $U\mathbf{x} = \mathbf{c}$ is a equivalent system of $A\mathbf{x} = \mathbf{b}$, in which $U = MA = E_k \cdots E_1 A$ and $\mathbf{c} = M\mathbf{b} = E_k \cdots E_1 \mathbf{b}$.

Elementary Matrices

- **Definition:**
  A sequence of special matrices $E_k, \cdots, E_1$ to obtain an equivalent system in row echelon form by matrix multiplications rather than row operations.
- Three types of elementary matrices corresponding to elementary row operations:
  - Interchanging two rows:
    For an $n \times n$ matrix,

$$E_1 = \begin{pmatrix} 0 & 1 & \cdots & 0 \\ 1 & 0 & \cdots & 0 \\ \vdots & & & \vdots \\ 0 & \cdots & \cdots & 1 \end{pmatrix}$$

    we can interchange the first row and the second row by $E_1 A$ and interchange the first column and the second column by AE_1. Other interchanging are similar.
  - Multiply a row by a nonzero constant:
    For an $n \times n$ matrix,

$$E_2 = \begin{pmatrix} c & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & & & \vdots \\ 0 & \cdots & \cdots & 1 \end{pmatrix}$$

    we can multiply the first row by c by $E_2 A$ and multiply the first column by c by AE_2. Others are similar. And the first type and the second type are symmetric. Since suppose the changed entries is symmetric about the diagonal.
  - Adding a multiple of one row to another row:
    For an $n \times n$ matrix,

$$E_3 = \begin{pmatrix} 1 & 0 & \cdots & c \\ 0 & 1 & \cdots & 0 \\ \vdots & & & \vdots \\ 0 & \cdots & \cdots & 1 \end{pmatrix}$$

we can add c times the last row to the first row by E_3A and add c times the first column to the last column by AE_3. Others are similar.
  - In general, suppose that E is an $n \times n$ elementary matrix, then for a $n \times r$ matrix A, EA has the same effect of performing the row operation on A. For a $m \times n$ matrix B, BE has the same effect of performing the column operation on B.
- **Theorem 1.5.1**
  If E is an elementary matrix, then E is nonsingular and E^{-1} is an elementary matrix of the same type.
  - For the type 1, the inverse matrix of E_{ij} will also be E_{ij} since we can interchange the same rows(columns) i, j again.
  - For the type 2, the inverse matrix of $E_i(k)$ will be $E_i\left(\frac{1}{k}\right)$ since we can transform into the initial matrix by multiplying the ith row(column) by $\frac{1}{k}$.
  - For the type 3, the inverse matrix of $E_{ij}(k)$ will be $E_{ij}(-k)$ since we can transform into the initial matrix by adding $-k$ times jth rows(columns) to ith rows(columns). For a particular transformation of type 3, we can take the opposite number of the nonzero entry that is not on the diagonal to get $E_{ij}(-k)$.
- **Row Equivalent:**
  - Definition:
    A matrix B is row equivalent to a matrix A if there exists a finite sequence $E_1, E_2, \cdots, E_k$ of elementary matrices such that

    $$B = E_k E_{k-1} \cdots E_1 A$$

  - Properties:
    - If A is row equivalent to B, then B is row equivalent to A.
    - If A is row equivalent to B and B is row equivalent to C, then A is row equivalent to C.
    - If two augmented matrix $(A|\mathbf{b})$ and $(B|\mathbf{c})$ are row equivalent, then $A\mathbf{x} = \mathbf{b}$ and $B\mathbf{x} = \mathbf{c}$ are equivalent systems.
- **Theorem 1.5.2**
  Let A be an $n \times n$ matrix, the following are equivalent:
  - A is nonsingular.
  - $A\mathbf{x} = \mathbf{0}$ has only the trivial solution $\mathbf{0}$.
  - A is row equivalent to I.
  - A can be expressed as the product of some elementary matrices.
- **Corollary 1.5.3**
  The system $A\mathbf{x} = \mathbf{b}$ of n linear equations in n unknowns has a unique solution if and only if the matrix A is nonsingular.
- **A Method for computing A^{-1}:**

- Augment A by I and perform the elementary row operations to transform A to I and then I will be transformed into A^{-1} since suppose that $EA = I$, then $E = A^{-1}$, thus $EI = A^{-1}$. The process is to transform the augmented matrix $(A|I)$ into $(I|A^{-1})$.
- Example: Compute A^{-1} if

$$A = \begin{pmatrix} 1 & 4 & 3 \\ -1 & -2 & 0 \\ 2 & 2 & 3 \end{pmatrix}$$

solution:

$$\left(\begin{array}{ccc|ccc} 1 & 4 & 3 & 1 & 0 & 0 \\ -1 & -2 & 0 & 0 & 1 & 0 \\ 2 & 2 & 3 & 0 & 0 & 1 \end{array}\right) \rightarrow \left(\begin{array}{ccc|ccc} 1 & 4 & 3 & 1 & 0 & 0 \\ 0 & 2 & 3 & 1 & 1 & 0 \\ 0 & -2 & 3 & 0 & 2 & 1 \end{array}\right)$$
$$\rightarrow \left(\begin{array}{ccc|ccc} 1 & 4 & 3 & 1 & 0 & 0 \\ 0 & 2 & 3 & 1 & 1 & 0 \\ 0 & 0 & 6 & 1 & 3 & 1 \end{array}\right) \rightarrow \left(\begin{array}{ccc|ccc} 1 & 4 & 0 & \frac{1}{2} & -\frac{3}{2} & -\frac{1}{2} \\ 0 & 2 & 0 & \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \\ 0 & 0 & 6 & 1 & 3 & 1 \end{array}\right)$$
$$\rightarrow \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ 0 & 2 & 0 & \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \\ 0 & 0 & 6 & 1 & 3 & 1 \end{array}\right) \rightarrow \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ 0 & 1 & 0 & \frac{1}{4} & -\frac{1}{4} & -\frac{1}{4} \\ 0 & 0 & 1 & \frac{1}{6} & \frac{1}{2} & \frac{1}{6} \end{array}\right)$$

thus,

$$A^{-1} = \begin{pmatrix} -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{4} & -\frac{1}{4} & -\frac{1}{4} \\ \frac{1}{6} & \frac{1}{2} & \frac{1}{6} \end{pmatrix}$$

Diagonal and Triangular Matrices

- Triangular Matrices:
  - upper triangular:
    For an $n \times n$ matrix A, $a_{ij} = 0$ for $i > j$.
  - lower triangular:
    For an $n \times n$ matrix A, $a_{ij} = 0$ for $i < j$.
  - triangular:
    It is either upper triangular or lower triangular.
  - Remarks: A triangular matrix may have 0's on the diagonal but for a [strict triangular form](), it must be upper triangular with nonzero diagonal entries.
- Diagonal Matrices:
  For an $n \times n$ matrix A, $a_{ij} = 0$ whenever $i \neq j$. And diagonal matrices is both upper triangular matrix and lower triangular matrix.

Triangular Factorization

- **Represent the elimination process in terms of matrix factorization(only operation 3):**
  For an $n \times n$ matrix A and E is the elementary matrix of A, if we call the resulting matrix U and set the matrix L as the inverse matrix of E, then can use

  $$A = LU$$

  to represent the process of reduction. And the factorization of the matrix A into a unit lower triangular matrix L times a strict upper triangular matrix U is often referred to as an LU factorization.
- The matrix L:
  The matrix L is lower triangular with 1's on the diagonal, called unit lower triangular. And if $i > j$, then l_{ij} is the multiple of the jth row subtract from the ith row during the reduction process.($i - l_{ij} \cdot j$)

Section 1.5 Exercises

- Question 13th
  For the first and second type elementary matrix, they are symmetric so $E^T = E$ and they are the same type elementary matrix. For the third matrix, the k in the $E_{ij}(k)$ will be symmetric about the diagonal. Thus E is to add k times ith to jth row and E^T is to add k times jth to ith row and they are the same type elementary matrix.
- Question 14th
  Since

  $$t_{ij} = \sum_{k=1}^{n} u_{ik} r_{kj}$$

  and when $i < j$,

  $$\begin{aligned} u_{i1} = u_{i2} = \cdots = u_{i,i-1} = 0 \\ r_{j+1,j} = r_{j+2,j} = \cdots = r_{nj} = 0 \end{aligned}$$

  then

  $$t_{ij} = \sum_{k=1}^{n} u_{ik} r_{kj} = u_{i1} r_{1j} + \cdots + u_{in} r_{nj} = 0$$

  thus the matrix T is upper triangular matrix. When $i = j$,

  $$t_{jj} = u_{j1} r_{1j} + \cdots + u_{jj} r_{jj} + \cdots + u_{jn} r_{nj} = u_{jj} r_{jj}, j = 1, \ldots, n.$$